

A Machine-Checked Itô Calculus for Brownian Motion

Raphael Coelho*

October 2026

Abstract

We develop the Itô calculus of Brownian motion, machine-checked in Lean 4 over Mathlib and the `BrownianMotion` package. On a bounded interval $[0, T]$ the Itô integral is built as a Hilbert-space isometry, from a predictable-rectangle π -system through the density of simple adapted processes. One identity, that the integral at time t is the conditional-expectation projection of its terminal value onto $\mathcal{F}_t$, makes it an L^2 martingale, continuous in L^2 , with its time-indexed isometry; a maximal inequality then gives it a modification with continuous paths, a local martingale on the half-line. Two theorems pin the operator down. Its range together with the constants is all of $L^2(\mathcal{F}_T^B)$, which is martingale representation, proved by totality of Doléans exponentials with no Malliavin calculus; and it satisfies the `BrownianMotion` package’s axiomatic characterisation of a stochastic integral. Itô’s formula for $f(t, B_t)$ is proved in L^2 for $f \in C^3$ whose derivatives grow at most exponentially, with the stochastic term the Itô integral of the named integrand $\partial_x f(s, B_s)$; for every $f \in C^3$, with no growth condition, as a continuous local martingale, by localization at exit times; and at a fixed time for Itô processes with bounded adapted continuous coefficients and $f \in C^2$ with bounded derivatives. To our knowledge these are the first machine-checked constructions of the Itô integral and proofs of Itô’s formula in any proof assistant. The boundary is explicit: the integrator is Brownian motion or an Itô integral against it, integrands are square-integrable, and quadratic variation is established in mean, not pathwise. The development is about 27,000 lines of Lean in 81 modules, sorry-free, with the axioms of each headline result pinned to Lean’s three standard axioms by a build-enforced gate.

1 INTRODUCTION

Stochastic calculus sits oddly in the formalization landscape. Its prerequisites (measure-theoretic probability, conditional expectation, L^p spaces, Gaussian measures) are well represented in the major proof-assistant libraries. Its consumers (mathematical finance, stochastic control, stochastic partial differential equations) are fields with strong correctness cultures that cite formal methods as an aspiration. Yet the calculus itself long remained unformalized. Martingales in discrete and continuous time were formalized in Isabelle/HOL by Keskin [10]; the central limit theorem was machine-checked a decade ago [1]; and in 2025 Degenne, Ledvinka, Marion and Pfaffelhuber completed the first formal construction of Brownian motion, in Lean 4 [5, 2]. Their package is the foundation this work builds on, and it is now growing a general theory of stochastic integration: an axiomatic characterisation of the stochastic integral, local martingales, and the Doob–Meyer

*Independent researcher. ORCID 0009-0001-6601-1023. Correspondence: raphaelrrcoelho@gmail.com. Artifact: <https://github.com/formal-applied-math/formal-mathfin>, release v1.5.0.

decomposition. The Itô integral of Brownian motion, Itô's formula and the martingale theory of the integral are the layer this paper supplies.

The development is the stochastic-calculus core of a larger library of formalized mathematical finance [4], in which it carries the continuous-time results: Girsanov's theorem, martingale representation and market completeness, stochastic differential equations, and the Black–Scholes pricing that rests on them; the library's no-arbitrage theory is treated separately [3]. Where the library overview surveys breadth, this paper is for readers who want to know exactly what was proved about the calculus, under which hypotheses, and by which proof architecture.

Setting. All results are stated over a probability space $(\Omega, \mathcal{F}, \mu)$ carrying a process $B: \mathbb{R}_{\geq 0} \rightarrow \Omega \rightarrow \mathbb{R}$ whose finite-dimensional laws are those of Brownian motion (the pre-Brownian interface `IsPreBrownianReal`, developed in the `BrownianMotion` package and now part of `Mathlib`), with measurable coordinates and, where stated, continuous paths. $(\mathcal{F}_t)$ is the natural filtration of B (`natFiltration`).

Contributions.

- (1) *The Itô integral as an isometry* (Section 5). The integrand space H_T is the L^2 space of the product measure on $(0, T] \times \Omega$ trimmed to the predictable σ -algebra; simple adapted processes are dense in it, and the Itô integral $\mathcal{I}_T$ (`itoIntegralCLM_T`) is the unique continuous linear extension of the elementary integral, an isometry.
- (2) *The integral as a process* (Section 6). The load-bearing identity $(\varphi \bullet B)_t = \mathbb{E}[\mathcal{I}_T(\varphi) \mid \mathcal{F}_t]$ in L^2 makes adaptedness, the martingale property, the contraction bound, the terminal and time-indexed Itô isometries and L^2 -continuity in t corollaries.
- (3) *Itô's formula in L^2* (Section 7). For $f(t, x)$ whose partial derivatives $f_t, f_x, f_{xx}, f_{tt}, f_{tx}, f_{xxx}$ are bounded, and then for those growing at most like $e^{\lambda|x|}$,

$$f(T, B_T) - f(0, B_0) = \mathcal{I}_T(f_x(\cdot, B)) + \int_0^T (f_t + \frac{1}{2}f_{xx})(s, B_s) ds \quad \mu\text{-a.s.},$$

with the integrand of $\mathcal{I}_T$ named as the class of $(s, \omega) \mapsto f_x(s, B_s(\omega))$, by a discrete-to-continuous argument through weighted quadratic variation with explicit L^2 remainder bounds and, for the growth case, a cutoff limit carried by the isometry.

- (4) *The pathwise integral* (Section 8). The integral process has an almost surely continuous modification whose everywhere-continuous representative is a local martingale for the null-augmented filtration, on $[0, T]$ and on all of $\mathbb{R}_{\geq 0}$.
- (5) *Itô's formula without growth conditions* (Section 9). For every f whose six partial derivatives above are continuous, with no bound of any kind, $f(t, B_t) - f(0, B_0) - \int_0^t (f_t + \frac{1}{2}f_{xx})(s, B_s) ds$ is a continuous local martingale, by localization at exit times.
- (6) *Itô processes* (Section 10). Quadratic variation of an Itô process with adapted volatility, Riemann–Stieltjes sums against an Itô integral, and Itô's formula for $X = x_0 + \int b ds + \int \sigma dB$ with bounded adapted continuous coefficients.
- (7) *The range and the uniqueness of the integral* (Section 11). Martingale representation: every square-integrable $\mathcal{F}_T^B$ -measurable F is $\mathbb{E}[F] + \mathcal{I}_T(\varphi)$ for a unique $\varphi \in H_T$. Integration against $M = \varphi \bullet B$ with its bracket. And agreement with the `BrownianMotion` package's axiomatic stochastic integral.

- (8) *What the calculus carries* (Section 12). Girsanov’s theorem for bounded predictable drifts, and existence and uniqueness for Lipschitz stochastic differential equations on short horizons.

Section 13 sets the boundary precisely: which parts of the classical theory are present, which are absent, and why the title says “a” calculus rather than “the” calculus.

Changes from the second version. Sections 9–12 are new, as are the growth-condition and process forms of Itô’s formula in Section 7. The bounded-derivative formula now names its integrand (the second version displayed an existentially quantified $g_{f'}$ whose prose described more than the statement said). For the same reason the keystone $\int_0^T B \, dB$ is now stated as what the Lean proves, with the named form derived from the growth-condition formula (Section 5.3). The pre-Brownian interface is `IsPreBrownianReal`, now in Mathlib; the predictable rectangles are consumed from the upstream package; and the artifact, pins and sizes are those of release v1.5.0. Smaller descriptions are corrected against the source: the σ -algebra the rectangles generate is identified here, not by an upstream lemma; the maximal inequality is Doob’s weak-type inequality applied to simple processes; and the closed form of $\int B \, dB$ preceded the general Riemann bridge rather than consuming it. The scope section is rewritten. Of the limitations it recorded, the restriction of Itô’s formula to bounded derivatives is lifted; the unrestricted formula, which the second version placed beyond the development in its C^2 form, is now proved for C^3 functions in local-martingale form, while the unrestricted C^2 formula for $f(t, B_t)$ remains open.

2 SETTING AND VERIFICATION DISCIPLINE

2.1 FOUNDATIONS CONSUMED

The development is built in Lean 4 [6] over Mathlib [11] and the `BrownianMotion` package [2, 5], whose construction of Brownian motion this work consumes rather than reproves; the artifact pins the package at commit 0d5b6eb. The pre-Brownian interface itself now lives in Mathlib. From the package the development takes the continuous modification of Brownian motion, the simple processes (`SimpleProcess`) and elementary predictable sets, the predictable rectangles of a filtration and the proof that they form a semiring of sets, the local-martingale predicate `IsLocalMartingale`, Doob’s maximal inequality, and the axiomatic stochastic integral of Section 11.3. Time is indexed by $\mathbb{R}_{\geq 0}$ (Lean’s `NNReal`); Lebesgue measure on time enters through `timeMeasure`, restricted to $(0, T]$.

2.2 AUDITABILITY AND REPRODUCIBILITY

Verification follows the library’s standing contract [4]. No `sorry` or `admit` occurs anywhere in the development. Two build-enforced audits run `#print axioms` under `#guard_msgs`: a curated one over headline results, and a generated one that pins every library constant a benchmark proof cites. Every numbered theorem of this paper, and every result it names as a headline, is pinned by one of them to `propext`, `Classical.choice` and `Quot.sound`, and a `sorry` or extra axiom beneath any of them fails the build. The `BrownianMotion` package has `sorry`s of its own at the pin; the audit is what certifies that none lies beneath a result reported here. A hash ledger records the exact source inputs and `toolchain` pins each benchmark entry was last verified under, so a result cannot go stale silently when a dependency changes. The statements displayed below are transcriptions of the Lean statements; where a hypothesis package is abbreviated in prose, the Lean declaration is the authority, and Appendix A maps every displayed result to its declaration, module and benchmark entry.

Role	Principal modules
Discrete engine	DiscreteIto, DiscreteItoPolynomial, ItoSquaringIdentity
Quadratic variation	QuadraticVariationL2, WeightedQuadraticVariation, BrownianQuadraticVariation, ItoProcessQV, AdaptedQuadraticVariation
Integral (isometry)	ItoIntegralL2, ItoIntegralCLM, ItoIsometryAdapted, ItoIntegralBrownian, ItoIntegralL2Dense, ItoIntegralCovariation, ItoIntegralLocality
Riemann bridges	ItoIntegralRiemannBridge, ItoIntegralRiemannBridgeTD, ItoIntegralRiemannBridgeAdapted, AdaptedProcessToLp
Integral as process	ItoIntegralProcess, ItoIntegralProcessGeneral, ItoIntegralProcessMartingale, ItoIntegralProcessIsometry
Itô's formula in L^2	ItoFormulaC2, ItoFormulaCLM, ItoFormulaRemainder, ItoFormulaSquaredL2, ItoFormulaTD, ItoFormulaTDRemainder, ItoFormulaProcess, ItoFormulaLocalized, ItoFormulaGBM
Pathwise regularity	ItoIntegralProcessContinuousModification, ItoIntegralProcessLocalMartingale, ItoIntegralProcessLocalMartingaleGeneral, ItoIntegralProcessL2Infinite, ItoIntegralProcessLocalMartingaleInfinite
Localization	ExitTime, ItoFormulaUnrestricted, ItoFormulaUnrestrictedLocMart
Itô processes	AdaptedStochasticIntegralFreezing, AdaptedRiemannStieltjes, ItoFormulaItoProcess, ItoFormulaAdapted, DriftProcessModification
Representation	MartingaleRepresentation, DoleansStepRepresentation, WienerExponentialTotality, BrownianCylinderGeneration
Integration against $\varphi \bullet B$	ItoIntegralAgainstMartingale, LpMulIsometry, PointwiseBracket, BracketCompensator, StochasticIntegralCharacterisation
Girsanov	ExpMartingaleQBrownian, GirsanovConstantTheta, GirsanovSimpleTheta, GirsanovAdaptedTheta, GirsanovPredictableTheta
SDEs	ItoProcessPredictable, SDEExistence, SDEPathwise, SDEUniqueness

TABLE 1. The Itô tower by role: principal modules, all under MathFin/Foundations/. The full tower, the import closure of these modules less the finance vocabulary, is 81 modules and about 27,000 lines.

2.3 SCALE

Counted as the import closure of its headline modules within the library, less the finance vocabulary that closure draws in, the Itô tower is 81 modules and about 27,000 lines of Lean, with some 830 theorems and lemmas; it lies downstream of Mathlib and BrownianMotion and upstream of the library's pricing layers. The 26 modules of the second version account for about 8,100 of those lines. [Table 1](#) groups the principal modules by role.

3 THE CLASSICAL THEORY

We recall the objects the formalization reproduces, in the form they take in the standard references [9, 8], so that the theorems below can be read against them.

Let $(B_t)_{t \geq 0}$ be a Brownian motion on $(\Omega, \mathcal{F}, \mu)$ with natural filtration $(\mathcal{F}_t)$. For a simple adapted process $\varphi_s = \sum_k \xi_k \mathbf{1}_{(t_k, t_{k+1}]}(s)$, with ξ_k bounded and $\mathcal{F}_{t_k}$ -measurable, the Itô integral is the finite sum $\int_0^T \varphi dB = \sum_k \xi_k (B_{t_{k+1}} - B_{t_k})$. The defining computation is the *Itô isometry*: because each ξ_k is independent of the forward increment and increments over

disjoint intervals are independent,

$$\mathbb{E} \left[\left(\sum_k \xi_k (B_{t_{k+1}} - B_{t_k}) \right)^2 \right] = \sum_k \mathbb{E}[\xi_k^2] (t_{k+1} - t_k) = \mathbb{E} \int_0^T \varphi_s^2 ds,$$

the cross terms vanishing by adaptedness and the diagonal contributing $\mathbb{E}[\xi_k^2] (t_{k+1} - t_k)$ because $\mathbb{E}[(B_{t_{k+1}} - B_{t_k})^2 | \mathcal{F}_{t_k}] = t_{k+1} - t_k$. The map $\varphi \mapsto \int_0^T \varphi dB$ is thus an isometry from the simple processes, carrying the $L^2(\lambda \otimes \mu)$ norm on the predictable side, into $L^2(\mu)$; as the simple processes are dense in the predictable L^2 space, it extends to a linear isometry on all of it (Section 5).

As a function of its upper limit, $t \mapsto \int_0^t \varphi dB$ is a continuous L^2 martingale with $\mathbb{E}[(\int_0^t \varphi dB)^2] = \mathbb{E} \int_0^t \varphi_s^2 ds$ (Sections 6 and 8). The quadratic variation of Brownian motion is $[B]_t = t$: along partitions of $[0, t]$ with mesh tending to zero, $\sum_k (B_{t_{k+1}} - B_{t_k})^2 \rightarrow t$. Itô's formula states that for $f \in C^{1,2}$

$$f(t, B_t) - f(0, B_0) = \int_0^t f_x(s, B_s) dB_s + \int_0^t (f_t + \frac{1}{2} f_{xx})(s, B_s) ds,$$

the second-order term surviving because $(dB)^2 = dt$ in the mean-square sense made exact by $[B]_t = t$. When $f_x(\cdot, B)$ is not square-integrable the stochastic integral is a local martingale, obtained by stopping at the exit times of compact sets (Section 9). Finally, the integral is onto: every centred square-integrable $\mathcal{F}_T^B$ -measurable random variable is $\int_0^T \varphi dB$ for a unique predictable φ , the martingale representation theorem (Section 11).

4 DESIGN OF THE FORMALIZATION

Four representation choices shape the development and are worth stating before the theorems that rest on them.

The integrand, after the upstream package. The simple integrands are not re-encoded here. `BrownianMotion` provides `SimpleProcess`, a bounded value at time zero together with finitely many bounded values on intervals, each measurable at the interval's left endpoint, and the predictable rectangles of a filtration together with the proof that they form a semiring of sets. This work consumes both: the elementary Itô integral is defined on `SimpleProcess`, the rectangle family is the upstream one, and the π -system property needed for Dynkin's theorem is projected from the upstream semiring. What is built here is the layer above: the identification of the σ -algebra the rectangles generate, the orthogonal-complement density and the isometric extension.

Time as `NNReal`. The time axis is $\mathbb{R}_{\geq 0}$, not $\mathbb{R}$ with a positivity side condition. Increments $B_t - B_s$ then carry $s \leq t$ in their type-level data rather than as a hypothesis to discharge, and the bounded horizon enters through a single finite measure (`timeMeasure_T`, of total mass T) rather than through interval hypotheses threaded across lemmas.

A trimmed measure. The integrand space is the L^2 space of a measure that has already been restricted to the predictable σ -algebra, rather than an ambient L^2 space with a sub- σ -algebra measurability side condition. Predictability is then a property of membership in H_T , not a hypothesis threaded through every limit: a Cauchy sequence in H_T has its limit in H_T , so when Itô's formula realizes $s \mapsto f'(B_s)$ as an L^2 limit of step processes the limit's predictability is free. The cost is a one-time investment in the trimmed-measure L^2 API; the saving recurs at every density and convergence argument downstream.

One construction pattern, used throughout. The integral, the integral as a process and the integral against an Itô integral are built by the same idiom: a norm-preserving linear map on a dense subspace, then its unique continuous linear extension (Mathlib's `LinearMap.extendOfNorm`). Every identity proved on simple processes then transfers to all of H_T by continuity and density, through the equalizer property of dense ranges (`DenseRange.equalizer`). That mechanism is what turns [Section 6](#) into a sequence of short arguments.

5 THE L^2 ITÔ INTEGRAL

5.1 THE INTEGRAND SPACE

Fix $T \in \mathbb{R}_{\geq 0}$ and write λ_T for Lebesgue measure on time restricted to $(0, T]$. The predictable rectangles

$$\mathcal{R} = \{(s, t] \times A : s < t, A \in \mathcal{F}_s\} \cup \{\{0\} \times A_0 : A_0 \in \mathcal{F}_0\}$$

of the natural filtration form a π -system (`isPiSystem_predictableRect`) that generates the predictable σ -algebra $\mathcal{P}$ (`generateFrom_predictableRect`, proved here directly); [Listing 1](#) shows the definition, which is now the upstream one. The integrand space is

$$H_T = L^2((0, T] \times \Omega, \mathcal{P}, (\lambda_T \otimes \mu) \upharpoonright_{\mathcal{P}}),$$

realized as the L^2 space of the product measure trimmed to $\mathcal{P}$ (`trimMeasure_T`), and the Itô integral is a continuous linear map out of it ([Listing 2](#)).

Listing 1. The predictable rectangles, consumed from `BrownianMotion` (`Foundations/ItoIntegralCLM`).

```
def predictableRect (hBmeas : ∀ t, Measurable (B t)) :
  Set (Set (ℝ≥0 × Ω)) :=
  (ItoIntegralL2.natFiltration (mΩ := mΩ) hBmeas).predictableRectangles

lemma isPiSystem_predictableRect (hBmeas : ∀ t, Measurable (B t)) :
  IsPiSystem (predictableRect (mΩ := mΩ) hBmeas) :=
  (MeasureTheory.Filtration.isSetSemiring_predictableRectangles _).isPiSystem
```

Listing 2. The Itô integral and its isometry (`Foundations/ItoIntegralCLM`).

```
noncomputable def itoIntegralCLM_T (hB : IsPreBrownianReal B μ) (T : ℝ≥0)
  (hBmeas : ∀ t, Measurable (B t)) :
  Lp ℝ 2 (trimMeasure_T (μ := μ) T hBmeas) → L[ℝ] Lp ℝ 2 μ :=
  (itoAssembly_T hB T hBmeas).extendOfNorm (simpleAssembly_T (μ := μ) T hBmeas)

theorem itoIntegralCLM_T_norm (hB : IsPreBrownianReal B μ) (T : ℝ≥0)
  (hBmeas : ∀ t, Measurable (B t))
  (f : Lp ℝ 2 (trimMeasure_T (μ := μ) T hBmeas)) :
  ||itoIntegralCLM_T hB T hBmeas f|| = ||f||
```

5.2 SIMPLE PROCESSES, DENSITY, ISOMETRY

A simple adapted process is a finite combination of indicator rectangles $\zeta_s \cdot \mathbf{1}_{(s, t]}$ with ζ_s bounded and $\mathcal{F}_s$ -measurable; `TBoundedSP` is the submodule of those supported in $[0, T]$ and `simpleProcessL2_T` sends it into H_T . The elementary integral maps such a process to $\sum \zeta_s (B_t - B_s)$ (`itoAssembly_T`); independence and the Gaussian increment moments give the elementary Itô isometry (`assembly_isometry_T`).

Density of the simple processes in H_T (`simpleAssembly_T_denseRange`) is the technical core of the construction and the part of it not inherited from the upstream package. It is proved by the orthogonal-complement method. Take $\psi \in H_T$ orthogonal to every elementary integrand. Testing against the indicator of a rectangle $(a, b] \times F$ shows that the integral of ψ over that rectangle vanishes; the measurable sets on which $\int \psi = 0$ form a λ -system containing the rectangle π -system, so Dynkin's theorem extends the vanishing to all of $\mathcal{P}$ (`setIntegral_eq_zero_of_orthogonal_pred`), forcing $\psi = 0$ almost everywhere. The Itô integral $\mathcal{I}_T$ (`itoIntegralCLM_T`) is then the unique isometric continuous linear extension of the elementary integral. The same density, run over the σ -finite time axis by exhaustion, gives the unbounded-horizon isometry on $L^2(\mathbb{R}_{\geq 0} \times \Omega)$ (`itoIntegralL2_norm`).

5.3 THE KEYSTONE EXAMPLE

The integral's first consumer was the closed form of $\int_0^T B \, dB$: there is an integrand $g_B \in H_T$, built as the H_T -limit of truncated left-endpoint step processes of $s \mapsto B_s$, with

$$\mathcal{I}_T(g_B) = \frac{1}{2}(B_T^2 - B_0^2 - T) \quad (\text{itoIntegralCLM_T_brownian}).$$

The Itô correction $-T/2$ emerges as follows: the summation-by-parts identity `discrete_squaring_identity` writes

$$\sum_k B_{s_k}(B_{s_{k+1}} - B_{s_k}) = \frac{1}{2} \left(B_T^2 - B_0^2 - \sum_k (B_{s_{k+1}} - B_{s_k})^2 \right),$$

whose left side is the discretized $\int_0^T B \, dB$ and whose squared-increment sum converges in L^2 to T ([Section 7.2](#)); what is left is $\frac{1}{2}(B_T^2 - B_0^2 - T)$. Because the left-endpoint coefficients B_{s_k} are unbounded Gaussians, the sums are not elementary integrals of simple processes; the proof truncates them to $[-M, M]$ and removes the truncation in the L^2 limit, which is how the Itô integral is defined for unbounded integrands in any case. The general Riemann bridge, which identifies $\mathcal{I}_T$ of a step-process realization of a bounded continuous function of B with the classical partial sums and feeds Itô's formula, came afterwards and builds on this file.

The statement records the integral, not the integrand. The construction makes g_B the class of B , but the theorem does not say so, and by martingale representation ([Theorem 11.1](#)) the bare existence of *some* integrand would hold for any centred square-integrable $\mathcal{F}_T^B$ -measurable target. The named form, $\mathcal{I}_T(B) = \frac{1}{2}(B_T^2 - B_0^2 - T)$ with B the class of $(s, \omega) \mapsto B_s(\omega)$, is the instance $f(t, x) = x^2$ of the growth-condition formula ([Theorem 7.3](#)), whose statement names its integrand: it gives g equal to $2B$ almost everywhere with $B_T^2 - B_0^2 = \mathcal{I}_T(g) + T$. That instance is a few lines from [Theorem 7.3](#) but is not stated separately in the library. The identity remains the smallest statement in which the calculus is visibly non-classical, and it anchors the remainder analysis of [Section 7](#).

5.4 DESIGN NOTE: TWO ISOMETRIES

The library separately maintains a Wiener-integral isometry for deterministic integrands. The two are deliberately distinct: the Wiener integral needs no filtration and serves the Gaussian-process and distributional layers, while $\mathcal{I}_T$ is the adapted object the martingale theory of [Section 6](#) requires. Collapsing them would shorten the code and damage the mathematics; the duplication is the design, and it is recorded as such in the library's dependency blueprint.

6 THE INTEGRAL AS A PROCESS

6.1 THE SIMPLE LAYER

For a simple process V and $t \leq T$, the partially summed elementary integral $(V \bullet B)_t$ is adapted (`itoSimpleProcess_adaptedAt`) and a martingale (`itoSimpleProcess_isMartingale`). The martingale step is the conditional vanishing of adapted-weighted future increments: by the characterizing property of conditional expectation it suffices that $\int_A \xi(B_{t_1} - B_{t_0}) d\mu = 0$ for every $A \in \mathcal{F}_{t_0}$, and $\xi \mathbf{1}_A$ is again bounded and $\mathcal{F}_{t_0}$ -measurable, so the increment, independent of $\mathcal{F}_{t_0}$ with mean zero, factors the integral as $\mathbb{E}[\xi \mathbf{1}_A] \mathbb{E}[B_{t_1} - B_{t_0}] = 0$. The martingale proof runs this argument set by set; the conditional statement it amounts to is recorded on its own in [Listing 3](#). The simple layer also carries the per-time isometry $\mathbb{E}[(V \bullet B)_t^2] = \int_0^t \mathbb{E}[V_s^2] ds$ (`itoSimpleProcess_isometry_time`) and L^2 -continuity in t (`itoSimpleProcessLp_l2_continuous`); the next subsection lifts all three to general integrands at once.

Listing 3. Adapted weights kill future increments (Foundations/ItoIntegralProcessMartingale).

```
theorem condExp_adapted_mul_increment (hBmeas : ∀ t, Measurable (B t))
  {t0 t1 : ℝ≥0} (ht : t0 ≤ t1) {φ : Ω → ℝ}
  (hφ : ItoIsometryAdapted.AdaptedAt B t0 φ) {C : ℝ} (hC : ∀ ω, |φ ω| ≤ C) :
  μ[fun ω ↦ φ ω * (B t1 ω - B t0 ω) | ItoIntegralL2.natFiltration hBmeas t0]
  =m[μ] 0
```

6.2 THE KEY IDENTITY

For general $\varphi \in H_T$ the process is a continuous linear map (`itoProcessCLM`, the extension of the simple layer), and the development's central structural fact is the following.

THEOREM 6.1 (`itoProcessCLM_eq_condExpL2`). *For every $t \geq 0$ and $\varphi \in H_T$, as elements of $L^2(\Omega, \mu)$,*

$$(\varphi \bullet B)_t = \mathbb{E}[\mathcal{I}_T(\varphi) \mid \mathcal{F}_t],$$

the right side realized by Mathlib's L^2 conditional-expectation projection `condExpL2`.

Listing 4. The integral process is the conditional projection of its terminal value (Foundations/ItoIntegralProcessGeneral).

```
theorem itoProcessCLM_eq_condExpL2 (T t : ℝ≥0) (hBmeas : ∀ u, Measurable (B u))
  (φ : Lp ℝ 2 (trimMeasure_T (μ := μ) T hBmeas)) :
  itoProcessCLM hB T t hBmeas φ
  = (condExpL2 ℝ ℝ ((natFiltration hBmeas).le t)
    (itoIntegralCLM_T hB T hBmeas φ) : Lp ℝ 2 μ)
```

The proof is two lines of mathematics: both sides are continuous linear in φ ; on the dense simple subspace they agree, because there the identity is the simple layer's martingale property; conclude by the equalizer property of dense ranges. Everything else about the process is a corollary of [Theorem 6.1](#) and standard properties of conditional expectation:

- *adaptedness*, since a `condExpL2` projection lands in the $\mathcal{F}_t$ -measurable subspace;
- the *martingale property* (`itoIntegralProcessGen_isMartingale`), by the tower property;
- the *contraction bound* $\|(\varphi \bullet B)_t\|_{L^2} \leq \|\varphi\|_{H_T}$ (`itoProcessCLM_norm_le`), since conditional expectation is a contraction;

- the *terminal isometry* at $t = T$ (`itoProcessCLM_norm_terminal`);
- L^2 -*continuity* of $t \mapsto (\varphi \bullet B)_t$ (`itoIntegralProcessGen_l2_continuous`), by a t -uniform contraction estimate against an approximating simple sequence.

Defining the process as a projection of its terminal value is what turns the martingale theory of the Itô integral into a sequence of corollaries rather than a sequence of constructions. Two further structural facts follow the same pattern. The isometry polarizes: $\mathbb{E}[\mathcal{I}_T(\varphi) \mathcal{I}_T(\psi)] = \int \varphi \psi d(\lambda_T \otimes \mu)$, the covariation of two Itô integrals (`ItoIntegralCovariation`). And the integral is local in time: if φ vanishes on $[0, a]$ and Z is $\mathcal{F}_a$ -measurable, then $\mathcal{I}_T(Z\varphi) = Z \mathcal{I}_T(\varphi)$ (`ItoIntegralLocality`), the “freeze the past, integrate the future” step behind the Doléans exponentials of [Section 11.1](#).

6.3 THE TIME-INDEXED ITÔ ISOMETRY

The terminal isometry is the classical $\|\mathcal{I}_T(\varphi)\|_{L^2}^2 = \|\varphi\|_{H_T}^2$. Its refinement to intermediate times, the energy law of the integral as a process, is the second fact every textbook calls “the Itô isometry”.

THEOREM 6.2 (`itoProcessCLM_norm_sq`). *For every $t \leq T$ and $\varphi \in H_T$,*

$$\mathbb{E}[(\varphi \bullet B)_t^2] = \int_{(0,t] \times \Omega} \varphi^2 d(\lambda_T \otimes \mu) \upharpoonright_{\mathcal{P}} = \int_0^t \mathbb{E}[\varphi_s^2] ds.$$

Listing 5. The time-indexed Itô isometry (Foundations/ItoIntegralProcessIsometry).

```
theorem itoProcessCLM_norm_sq {t T : ℝ≥0} (htT : t ≤ T) (hBmeas : ∀ u, Measurable
  ↪ (B u))
  (φ : Lp ℝ 2 (trimMeasure_T (μ := μ) T hBmeas)) :
  ||itoProcessCLM hB T t hBmeas φ|| ^ 2
  = ∫ z in (Set.Ioc 0 t ×s (Set.univ : Set Ω)), (φ z) ^ 2
    ∂(trimMeasure_T (μ := μ) T hBmeas)
```

The proof is again the equalizer pattern, against a different pair of continuous functionals: $\varphi \mapsto \|(\varphi \bullet B)_t\|^2$ and $\varphi \mapsto \int_{(0,t]} \varphi^2$ agree on simple processes by a band reconciliation that splits the trimmed measure across $(0, t] \cup (t, T]$ (`itoSimpleProcessLp_band_isometry`, `integral_rectTerm_mul_band`). Stating it for general integrands at every t , not only at $t = T$ and not only for simple processes, is what makes the integral as a process a complete object rather than a martingale with a missing variance law.

7 ITÔ’S FORMULA IN L^2

7.1 ARCHITECTURE

The route to Itô’s formula is discrete-to-continuous ([Figure 1](#)).

1. *Discrete identity.* For any partition, second-order Taylor expansion of f along the increments gives an exact discrete Itô identity (`DiscreteIto`) with an explicit Lagrange remainder.
2. *Weighted quadratic variation.* The term $\sum_k f''(B_{t_k})(\Delta_k B)^2$ converges in L^2 to $\int_0^T f''(B_s) ds$; this is proved for adapted weight processes (`WeightedQuadraticVariation`), reusing the Gaussian moment estimates of the unweighted limit.
3. *Remainder.* The third-order remainder is $O(1/n)$ in L^2 under a bound on f''' (`ItoFormulaRemainder`); the time-dependent variant needs a two-dimensional remainder with mixed partials, also $O(1/n)$ (`ItoFormulaTDRemainder`).

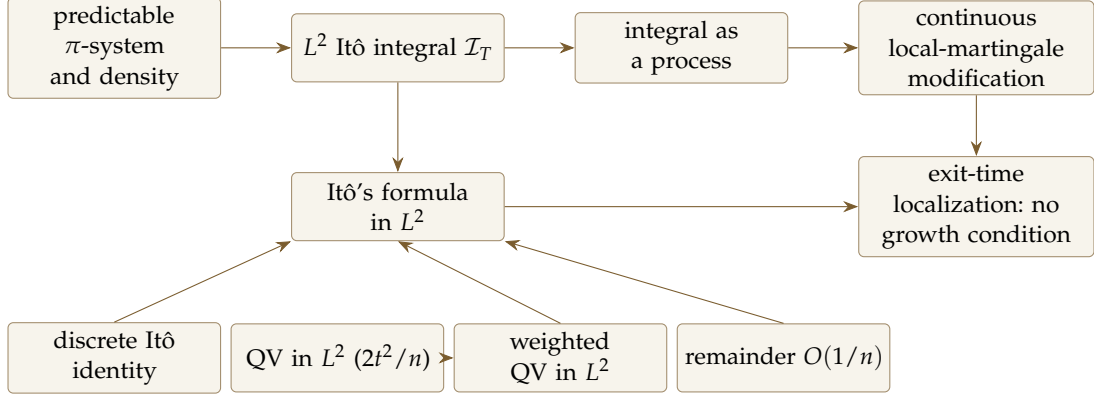


FIGURE 1. Proof architecture. The isometry layer (top) builds the integral, the process and its continuous modification; the discrete-to-continuous layer (bottom) supplies the vanishing terms, which meet at Itô's formula; localization combines the formula with the pathwise process to remove every growth condition.

4. *Riemann bridge*. The discretized stochastic term is $\mathcal{I}_T$ applied to a step-process realization of $f'(B)$.
5. *Limit*. The step integrands converge in H_T to the class of $s \mapsto f'(B_s)$. Predictability of the limit is free (the limit is taken inside H_T); its existence for bounded continuous data is `itoIntegralCLM_T_of_bdd_cont`, a dominated-convergence argument on the trimmed measure.

7.2 THE FLUCTUATION ENGINE: QUADRATIC VARIATION IN L^2

The second-order term of Itô's formula is deterministic in the limit for one reason: the squared increments of Brownian motion have vanishing fluctuation in L^2 . Along the uniform partition of $[0, t]$ into n pieces of width $\Delta = t/n$,

$$\left\| \sum_k (B_{s_{k+1}} - B_{s_k})^2 - t \right\|_{L^2(\mu)}^2 = \frac{2t^2}{n} \longrightarrow 0 \quad (\text{tendsto_qv}).$$

The value is exact, not asymptotic: for any partition $0 = s_0 \leq \dots \leq s_n$ the mean-square error is $\sum_k 2(s_{k+1} - s_k)^2$ (`sum_increment_sq_sub_sq_integral`), which is $2t^2/n$ for the uniform one. Writing $Y_k = (B_{s_{k+1}} - B_{s_k})^2 - \Delta$ for the centred squared increment, the Y_k over disjoint intervals are independent, so the cross terms vanish and $\mathbb{E}[(\sum_k Y_k)^2] = \sum_k \mathbb{E}[Y_k^2]$. Each term is the Gaussian kurtosis, $\mathbb{E}[Y_k^2] = \mathbb{E}[(B_{s_{k+1}} - B_{s_k})^4] - \Delta^2 = 3\Delta^2 - \Delta^2 = 2\Delta^2$, using $\mathbb{E}[X^4] = 3(\text{Var } X)^2$ for a centred Gaussian (`integral_pow4_gaussianReal`, transported to the increment law by `integral_increment_pow4`). Summing, $n \cdot 2\Delta^2 = 2t^2/n$. The fourth-moment constant 3, and nothing else, is the reason the quadratic variation is t rather than 0; this is the formal content of $(dB)^2 = dt$.

7.3 THE WEIGHTED SECOND-ORDER TERM

Itô's formula needs this limit carrying a weight: for a bounded adapted process w with continuous paths,

$$\sum_k w_{t_k} (B_{t_{k+1}} - B_{t_k})^2 \longrightarrow \int_0^T w_s ds \quad \text{in } L^2 \quad (\text{tendsto_weighted_qv_process}),$$

instantiated at $w_s = f''(B_s)$ for the autonomous formula and $w_s = f_{xx}(s, B_s)$ for the time-dependent one. The proof splits the difference into a fluctuation term and a Riemann term.

The fluctuation term $\sum_k w_{t_k}((B_{t_{k+1}} - B_{t_k})^2 - \Delta_k)$ vanishes in L^2 by the orthogonality-plus-kurtosis estimate above, now carrying the $\mathcal{F}_{t_k}$ -measurable bounded weight: adaptedness of w_{t_k} is exactly what keeps the cross terms zero (integral_adapted_mul_centered_sq). The Riemann term vanishes pathwise by continuity of $s \mapsto w_s(\omega)$, then in L^2 by dominated convergence against the bound on w . Neither half uses where the weight came from, only that it is adapted, bounded and path-continuous, which is why one lemma serves the autonomous, the time-dependent, and later the Itô-process formula.

7.4 THE REMAINDER

What survives after the first- and second-order terms is a sum of cubic Taylor remainders, and it must vanish in L^2 . The per-step remainder R_k obeys $|R_k| \leq C_3 |B_{s_{k+1}} - B_{s_k}|^3$ under $|f'''| \leq C_3$ (abs_discreteTaylorRemainder_1e). Hence $\mathbb{E}[R_k^2] = O(\mathbb{E}[(\Delta_k B)^6]) = O(\Delta^3)$ by the Gaussian sixth moment (integral_pow6_gaussianReal), and Cauchy-Schwarz across the n terms gives

$$\mathbb{E}\left[\left(\sum_k R_k\right)^2\right] \leq n \sum_k \mathbb{E}[R_k^2] = O(n \cdot n \cdot (T/n)^3) = O(1/n) \longrightarrow 0.$$

The third order is the only place a third derivative is used. The time-dependent variant runs the same estimate on a two-dimensional Taylor remainder with mixed partials.

7.5 THE BOUNDED-DERIVATIVE FORMULA

THEOREM 7.1 (ito_formula_L2_bddDeriv). *Let B have measurable coordinates and continuous paths, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be three times differentiable with $|f'| \leq C_1$, $|f''| \leq C_2$, $|f'''| \leq C_3$. Then there is $g \in H_T$ with $g = f'(B_s(\omega))$ for $(\lambda_T \otimes \mu) \upharpoonright_{\mathcal{P}}$ -almost every (s, ω) , such that μ -almost surely*

$$f(B_T) - f(B_0) = \mathcal{I}_T(g) + \frac{1}{2} \int_{(0,T]} f''(B_s) ds.$$

Listing 6. Itô's formula, bounded-derivative class (Foundations/ItoFormulaCLM).

```

theorem ito_formula_L2_bddDeriv
  (hBmeas : ∀ t, Measurable (B t)) (hBcont : ∀ ω, Continuous (fun s : ℝ≥0 ↦ B s
    ↪ ω))
  (T : ℝ≥0) {f f' f'' f''' : ℝ → ℝ}
  (hf : ∀ x, HasDerivAt f (f' x) x) (hf' : ∀ x, HasDerivAt f' (f'' x) x)
  (hf'' : ∀ x, HasDerivAt f'' (f''' x) x)
  {C1 : ℝ} (hf1 : ∀ x, |f' x| ≤ C1) {C2 : ℝ} (hf2 : ∀ x, |f'' x| ≤ C2)
  {C3 : ℝ} (hf3 : ∀ x, |f''' x| ≤ C3) :
  ∃ gf' : Lp ℝ 2 (trimMeasure_T (μ := μ) T hBmeas),
    (↑gf' =m[trimMeasure_T (μ := μ) T hBmeas] fun z ↦ f' (B z.1 z.2)) ∧
    (fun ω ↦ f (B T ω) - f (B 0 ω)) =m[μ]
      (fun ω ↦ (itoIntegralCLM_T hB T hBmeas gf') ω
        + (1 / 2) * ∫ s in Set.Ioc 0 T, f'' (B s ω) ∂ItoIntegralL2.timeMeasure)

```

The first conjunct is what makes the second Itô's *formula* rather than a decomposition into some Itô integral: the stochastic term is the integral of the class of $f'(B)$, not of an unnamed element of H_T .

Proof sketch. Fix the uniform partition of $[0, T]$ into n pieces and expand f to second order along each increment. The telescoping sum gives the exact discrete identity $f(B_T) - f(B_0) = S_n + \frac{1}{2}Q_n + R_n$, with $S_n = \sum_k f'(B_{s_k})(B_{s_{k+1}} - B_{s_k})$, $Q_n = \sum_k f''(B_{s_k})(B_{s_{k+1}} - B_{s_k})^2$

and R_n the cubic remainder. Each term has a limit: $S_n \rightarrow \mathcal{I}_T(g)$ in L^2 because the step integrands converge to g in H_T and $\mathcal{I}_T$ is isometric; $Q_n \rightarrow \int_0^T f''(B_s) ds$ by the weighted quadratic variation; and $R_n \rightarrow 0$ by the remainder estimate. The elementary inequality $(a + b + c)^2 \leq 3a^2 + 3b^2 + 3c^2$ sends the total error to zero in L^2 , so the two sides agree in L^2 , hence almost surely. $\square$

Path continuity is supplied by the package's continuous modification and is genuinely used: the Riemann limits run along refining partitions evaluated on paths.

7.6 THE TIME-DEPENDENT FORMULA, AND THE FORMULA AS A PROCESS

THEOREM 7.2 (`ito_formula_td_L2_bddDeriv`). *Let $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ have everywhere partial derivatives $f_t, f_x, f_{xx}, f_{tt}, f_{tx}, f_{xxx}$, all bounded, with f_x and f_{xx} jointly continuous. Then there is $g \in H_T$ equal to $f_x(s, B_s(\omega))$ almost everywhere such that, μ -almost surely,*

$$f(T, B_T) - f(0, B_0) = \mathcal{I}_T(g) + \int_{(0,T]} (f_t + \frac{1}{2}f_{xx})(s, B_s) ds.$$

Two formalization points. First, joint continuity of f_t is not assumed: it is *derived* from the bounded mixed partials (`continuous_uncurry_of_bdd_partials`), keeping the hypothesis package at what a user can check. Second, the time-direction Taylor terms are handled by a boundary-cancellation decomposition in which three telescoping error sums vanish separately, glued by the three-term Cauchy–Schwarz estimate.

The formula also holds as a process. Under the same hypotheses there is an integrand F on the half-line such that, for every $t \leq T$, $f(t, B_t) - f(0, B_0) = (F \bullet B)_t + \int_0^t (f_t + \frac{1}{2}f_{xx})(s, B_s) ds$ almost surely, the integral process being a martingale with a continuous local-martingale modification (`ito_formula_td_process`). This statement, unlike [Theorem 7.2](#), does not name F .

7.7 EXPONENTIAL GROWTH

Bounded derivatives exclude the textbook's first examples: x^2 has an unbounded first derivative, and the geometric Brownian motion value function $S_0 \exp((r - \sigma^2/2)t + \sigma x)$ has all its derivatives proportional to $e^{\sigma x}$. The formula is lifted to them by a cutoff localization that reuses the bounded engine.

THEOREM 7.3 (`ito_formula_td_localized`). *The conclusion of [Theorem 7.2](#) holds when the six partial derivatives, instead of being bounded, satisfy $|f_\bullet(t, x)| \leq Ce^{\lambda|x|}$ for some C and $\lambda \geq 0$, and f_t, f_x, f_{xx} are jointly continuous.*

A smooth truncation ϕ_n , equal to the identity on $[-n, n]$ with bounded derivatives, makes $f_n(t, x) = f(t, \phi_n(x))$ satisfy the bounded hypotheses, so [Theorem 7.2](#) applies to each f_n , with integrands g_n . The boundary and drift terms converge in $L^2(\mu)$ by dominated convergence, the exponential dominators being integrable because Brownian marginals have all exponential moments. Hence the integrals $\mathcal{I}_T(g_n)$ converge and are Cauchy; the isometry transfers Cauchy-ness back to the g_n , which converge in the complete space H_T to some g ; continuity of $\mathcal{I}_T$ identifies the limit, and the almost-everywhere identification of g with $f_x(s, B_s)$ passes to the limit along a subsequence. The truncation is the antiderivative of a smooth bump, so every derivative bound comes from Mathlib rather than from explicit calculus. Applied to the exponential, the theorem gives the Itô decomposition of geometric Brownian motion (`ito_formula_gbm`) and, for the discounted price, the representation $e^{-rt}S_t = S_0 + \mathcal{I}_t(\sigma e^{-r\cdot} S)$ with its integrand named (`discountedGBM_eq_itoIntegral`), which is where the library's Black–Scholes layer meets the Itô integral.

7.8 SUPPORTING RESULTS

Quadratic variation. For a process with measurable coordinates and centred Gaussian increments of variance $t - s$ (neither independence of increments nor path continuity is assumed), the expected squared-increment sums along equipartitions of $[0, t]$ converge to t (`qv_equals_t`); the L^2 form along uniform partitions is `tendsto_qv`, and convergence in probability is `tendstoInMeasure_qv`. The L^1 statement needs only the second moment of the increments, and so strictly strengthens the classical claim.

Expectation-level Itô. For f with f, f' and f'' bounded and f'' continuous, $\mathbb{E}[f(B_t)] = f(0) + \frac{1}{2} \int_0^t \mathbb{E}[f''(B_s)] ds$ for $t > 0$ (`expectation_ito`). It is the bridge between the stochastic tower of this paper and the heat-kernel tower of the parent library, where the Black–Scholes equation is derived through Feynman–Kac.

7.9 NOTES ON THE PROOF ENGINEERING

The density argument is the only place a monotone-class induction was unavoidable in the construction of the integral; expressing it against a trimmed measure, so that the inductively built sets live in the right σ -algebra by construction, is what kept it tractable. The dominated-convergence step that produces the integrand of Itô’s formula is delicate because the dominating function must itself be predictable; the trimmed-measure design is what makes the domination land inside H_T without a separate measurability obligation.

8 FROM L^2 TO PATHWISE: THE CONTINUOUS LOCAL MARTINGALE

Sections 6 and 7 are an L^2 theory: the process $t \mapsto (\varphi \bullet B)_t$ is continuous *as a curve in* $L^2(\mu)$, and its laws are identities of L^2 elements. The classical object is sharper: for almost every ω it has a continuous path, and it is a continuous local martingale in the pathwise sense. This section builds that object, first on $[0, T]$ and then on the half-line.

8.1 THE PATHWISE GAP AND THE MAXIMAL INEQUALITY

L^2 -continuity in t does not by itself give a continuous path: it controls each time separately, not the trajectory. The bridge is a maximal inequality. `BrownianMotion` supplies Doob’s weak-type maximal inequality for right-continuous martingales; applied to the simple integral processes, with the L^1 norm bounded by the L^2 norm and the isometry, it gives $\mu\{\sup_{t \leq T} |(V \bullet B)_t| \geq \varepsilon\} \leq \varepsilon^{-1} \|V\|_{H_T}$ (`itoSimpleProcess_maximal_prob`). A geometric subsequence makes these tails summable, and Borel–Cantelli makes the approximating step-process integrals converge uniformly on $[0, T]$ for almost every ω . A uniform limit of continuous functions is continuous, so the limit path is continuous off a null set.

8.2 THE CONTINUOUS MODIFICATION

THEOREM 8.1 (`exists_continuous_modification_itoProcess`). *Let B have every path continuous. For every $\varphi \in H_T$ there is a process $\tilde{X}: \mathbb{R}_{\geq 0} \times \Omega \rightarrow \mathbb{R}$ with μ -almost surely continuous paths such that $\tilde{X}_t = (\varphi \bullet B)_t$ in $L^2(\mu)$ for every $t \leq T$.*

The representative `itoContinuousMod` is the pointwise limit of the approximating step integrals on the good set where they converge uniformly. It equals $(\varphi \bullet B)_t$ almost surely at each t by uniqueness of limits in measure, and its paths are continuous by the uniform-convergence theorem; the supporting bound on the running supremum over $[0, t]$ is `norm_le_iSup_Iic`.

8.3 THE LOCAL MARTINGALE ON THE NULL-AUGMENTED FILTRATION

A modification fixes the paths but can break adaptedness, since the continuous representative is defined through a limit taken off a null set. The remedy is to complete the filtration: adjoining the μ -null sets to each $\mathcal{F}_t$ gives $\mathcal{F}_t^\mu = \mathcal{F}_t \vee \mathcal{N}$ (`augFiltration`). Completeness is all that is added; right-continuity, the other half of the usual conditions, is neither added nor needed.

THEOREM 8.2 (`exists_continuous_localMartingale_modification`). *Let B have every path continuous. For every $\varphi \in H_T$ there is a process X , adapted to $(\mathcal{F}_t^\mu)$ and with everywhere-continuous paths, such that $X_t = (\varphi \bullet B)_t$ in $L^2(\mu)$ for each $t \leq T$, and X is a local martingale for $(\mathcal{F}_t^\mu)$.*

Two points fix the content. The underlying L^2 process is a *true* martingale ([Section 6](#)); what the theorem adds is an everywhere-continuous representative, in the form the local-martingale interface of `BrownianMotion` takes, whose requirement of paths continuous for *every* ω is what forces an everywhere-defined representative. And the martingale property survives the completion, because a conditional expectation is unchanged when the filtration is enlarged by null sets (`condExp_sup_nulls`).

8.4 THE UNBOUNDED HORIZON

The bounded-horizon representatives are consistent: for $m \leq n$ the horizon- m and horizon- n continuous processes agree, almost surely and pathwise, on $[0, m)$, because both are continuous and both modify the same L^2 process there. Gluing them gives a single process on the half-line, reading at time t the horizon $\lceil t \rceil + 1$.

THEOREM 8.3 (`exists_continuous_localMartingale_modification_infinite`). *For every φ in the unbounded-horizon integrand space there is a process X with everywhere-continuous paths such that $X_t = (\varphi \bullet B)_t$ in $L^2(\mu)$ for every t , and X is a local martingale for $(\mathcal{F}_t^\mu)$ on $\mathbb{R}_{\geq 0}$.*

This is the Itô integral in the form the classical theory gives it: a continuous local martingale on the half-line. The glued process is in fact a true martingale with continuous paths (`exists_continuous_martingale_modification_infinite`), which is what the localization of [Section 9](#) uses.

9 ITÔ'S FORMULA WITHOUT GROWTH CONDITIONS

The L^2 formulas of [Section 7](#) need the integrand $f_x(\cdot, B)$ to lie in H_T , which is what the derivative bounds and growth bounds secure. Without them the stochastic term is no longer an L^2 integral, and the classical statement is that the compensated process is a continuous local martingale. That is the statement proved here.

THEOREM 9.1 (`ito_formula_unrestricted`). *Let B have measurable coordinates and continuous paths, and let $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ have everywhere partial derivatives $f_t, f_x, f_{xx}, f_{tt}, f_{tx}, f_{xxx}$ which, together with f , are jointly continuous. No growth or boundedness condition is imposed. Then there is a process M with every path continuous, which is a local martingale for the null-augmented filtration $(\mathcal{F}_t^\mu)$, such that for every $t \geq 0$, almost surely,*

$$f(t, B_t) - f(0, B_0) = M_t + \int_{(0,t]} (f_t + \frac{1}{2}f_{xx})(s, B_s) \, ds.$$

Listing 7. Itô's formula without growth conditions (Foundations/ItoFormulaUnrestrictedLocMart); five of the six derivative hypotheses and five of the seven continuity hypotheses are abbreviated.

```

theorem ito_formula_unrestricted (hB : IsPreBrownianReal B μ)
  (hBmeas : ∀ t, Measurable (B t)) (hBcont : ∀ ω, Continuous fun s : ℝ≥0 ↦ B s ω
  ↪ )
  {f f_t f_x f_xx f_tt f_tx f_xxx : ℝ → ℝ → ℝ}
  (hf_t : ∀ t x, HasDerivAt (fun s ↦ f s x) (f_t t x) t)
  -- ... hf_tt, hf_tx, hf_x, hf_xx, hf_xxx: the other five partial derivatives
  (hf_cont : Continuous fun p : ℝ × ℝ ↦ f p.1 p.2)
  -- ... joint continuity of f_t, f_x, f_xx, f_tt, f_tx
  (hf_xxx_cont : Continuous fun p : ℝ × ℝ ↦ f_xxx p.1 p.2) :
  ∃ M : ℝ≥0 → Ω → ℝ,
    (∀ ω, Continuous fun t ↦ M t ω) ∧
    IsLocalMartingale M (augFiltration (μ := μ) hBmeas) μ ∧
    (∀ t : ℝ≥0, (fun ω ↦ f (t : ℝ) (B t ω) - f 0 (B 0 ω)) =m[μ]
      (fun ω ↦ M t ω + ∫ s in Set.Ioc 0 t,
        (f_t (s : ℝ) (B s ω) + (1 / 2) * f_xx (s : ℝ) (B s ω)) ∂timeMeasure))

```

9.1 EXIT TIMES AS LOCALIZERS

The localizing sequence is the exit time of the closed exterior $\{|x| \geq N\}$, $\tau_N = \inf\{t : |B_t| \geq N\}$ (`exitTime`). Because the hit set is closed and the paths continuous, $\{\tau_N \leq i\}$ is the event $\{\exists s \in [0, i] : |B_s| \geq N\}$, which is measurable for $\mathcal{F}_i$ through a countable rational description. So τ_N is a stopping time for the raw natural filtration, with no right-continuity required; the open exterior $\{|x| > N\}$ would characterise only $\{\tau_N < i\}$, an event of $\mathcal{F}_{i+}$. The package's own localizing sequence of hitting times needs a complete and right-continuous filtration, which the natural filtration is not; the exit time does not. For continuous paths the exit times escape to infinity, since each path is bounded on compacts.

9.2 DOUBLE TRUNCATION AND AGREEMENT ON STOCHASTIC INTERVALS

Localizing in space is not enough: a general f has t -derivatives unbounded over $t \in \mathbb{R}$. The proof therefore truncates both arguments, $f^N(t, x) = f(\text{cut}_N(t), \text{cut}_N(x))$, with a smooth cutoff onto a compact square on which the continuous partials of f are bounded. The chain rule gives the six partials of f^N in closed form, each globally bounded, so the process form of the bounded formula applies and yields a truncated Itô identity with a continuous *true*-martingale modification M^N on the half-line. On the stochastic interval $\{t \leq \sigma_N\}$ with $\sigma_N = \min(\tau_N, N)$ both cuts are inert ($t \leq N$ and $|B_s| \leq N$ for $s \leq t$), so $f^N = f$ there and $M_t^N = M_t$ almost surely on $\{t \leq \sigma_N\}$. Capping the localizer at N is what keeps the time cut inert exactly where the agreement is used.

9.3 FROM PER-TIME AGREEMENT TO A LOCAL MARTINGALE

Agreement at each fixed t is lifted to agreement of whole paths on the stochastic interval, almost surely, by continuity and a countable dense set of times (`indistinguishable_on_stochInterval`). The compensated process M is adapted, because its drift primitive $\int_0^t (f_t + \frac{1}{2}f_{xx})(s, B_s) ds$ is $\mathcal{F}_t$ -measurable: after clamping the integrand's time to $[0, t]$ the integrand is a Carathéodory function (continuous in s , $\mathcal{F}_t$ -measurable in ω), hence jointly measurable, and its integral is measurable (`StronglyMeasurable.integral_prod_right`). With M adapted, the stopped indicator processes are martingales, and they assemble into the package's `IsLocalMartingale` with localizing sequence σ_N .

9.4 WHAT THE THEOREM SAYS, AND WHAT IT DOES NOT

Theorem 9.1 removes the growth conditions of [Section 7](#) entirely, but it says less than the classical formula in one respect: M is not identified as a stochastic integral $\int_0^t f_x(s, B_s) dB_s$, because the library's integral is defined for square-integrable integrands and $f_x(\cdot, B)$ need not be one. Where it is, the identification is [Theorem 7.3](#). The theorem covers every C^3 function, including those that grow faster than any exponential: applied to $f(t, x) = e^{x^2}$, for instance, it gives a continuous local martingale with no integrability of $f(t, B_t)$ assumed. The smoothness hypothesis is the one the bounded engine needs; the $C^{1,2}$ statement would require an additional approximation of f and is not formalized.

10 ITÔ PROCESSES

Every model in continuous-time finance is an Itô process $X = x_0 + \int b ds + \int \sigma dB$, so the formula must reach functions of X , not only of (t, B_t) . Three results lead there.

Quadratic variation of an Itô process. For $X = X_0 + A + \sigma \bullet B$ with a drift path A Lipschitz in time uniformly in ω , the squared-increment sums along the uniform partition converge to $\int_0^T \sigma_s^2 ds$: in L^2 for constant σ (ItoProcessQV), and in L^1 for σ bounded, adapted and continuous in time, with the weighted form $\sum_k w(t_k)(\Delta X_k)^2 \rightarrow \int_0^T w \sigma^2 ds$ for bounded adapted weights with almost surely continuous paths. The drift contributes nothing. The proof splits $(\Delta X_k)^2$ around the frozen increment $\sigma(t_k)\Delta B_k$: the freezing defects $D_k = \Delta M_k - \sigma(t_k)\Delta B_k$ of $M = \sigma \bullet B$ are orthogonal Itô integrals whose sum is the error of the frozen Riemann sum, so $\sum_k \|D_k\|^2 \rightarrow 0$ (AdaptedStochasticIntegralFreezing), and what remains is the weighted quadratic variation of B . The L^1 mode is a limitation of the method: an L^2 bound on the freezing part would need fourth moments of $\sigma \bullet B$.

Riemann–Stieltjes sums against an Itô integral. For $M = \varphi \bullet B$ and a bounded predictable weight w whose paths are almost surely continuous on $[0, T]$, the sums $\sum_k w(t_k)(M_{t_{k+1}} - M_{t_k})$ converge in mean square to the integral of w against M ([Section 11.2](#)), which is the Itô integral against B of the class of φw (AdaptedRiemannStieltjes). The route goes through $L^2(\langle M \rangle)$: each sum is the integral against M of a left-endpoint step process, the step processes converge to w by dominated convergence, and the isometry against M carries the convergence over. No path regularity of φ is needed.

Constant coefficients. For $X_t = X_0 + bt + \sigma B_t$ and f whose first three derivatives grow at most exponentially, the formula $f(X_T) - f(X_0) = \mathcal{I}_T(\sigma f'(X)) + \int_0^T (f'(X_s)b + \frac{1}{2}f''(X_s)\sigma^2) ds$ holds with the integrand named (ito_formula_itoProcess); this is the form the geometric Brownian motion needs.

THEOREM 10.1 (ito_formula_adapted). *Let B have every path continuous, let σ and b be bounded, adapted to the natural filtration of B , with every path continuous, and let $X = x_0 + \int_0^\cdot b ds + \int_0^\cdot \sigma dB$. For f twice differentiable with f' , f'' bounded and f'' continuous, almost surely*

$$f(X_T) - f(x_0) = \mathcal{I}_T(f'(X)\sigma) + \int_0^T \left(f'(X_s)b_s + \frac{1}{2}f''(X_s)\sigma_s^2 \right) ds,$$

the stochastic integral being the Itô integral at horizon T of the class of the bounded predictable process $f'(X)\sigma$.

The process X is defined (adaptedItoProcess) as x_0 plus the pathwise drift integral plus the continuous modification of the Itô integral process of σ . The proof applies the discrete Itô formula on the uniform partition, which splits $f(X_T) - f(x_0)$ into drift sums,

Riemann–Stieltjes sums against $M = \sigma \bullet B$, weighted quadratic variation and a remainder. The three random limits hold in measure; a common subsequence along which all three converge almost surely is extracted, and on such a path, where X is also continuous, a deterministic lemma (`ito_formula_of_tendsto`) passes the discrete identity to the limit. Writing the stochastic term as an integral against M and then, by the chain rule of [Section 11.2](#), against B gives the displayed form. Because the limit is taken almost surely along a subsequence rather than in L^2 at a rate, the Taylor remainder is controlled by the modulus of continuity of f'' times the quadratic variation, and no third derivative is needed: this is the library’s first Itô formula for C^2 functions, as well as its first for a process that is not a function of (t, B_t) . Its scope is recorded in [Section 13](#).

11 THE INTEGRAL AS AN OPERATOR: ITS RANGE AND ITS UNIQUENESS

Two theorems characterise $\mathcal{I}_T$ as an operator. Its range is as large as it can be, and it is the only reasonable integral.

11.1 MARTINGALE REPRESENTATION

THEOREM 11.1 (`exists_itoIntegral_representation`). *Let B have every path continuous. For every $F \in L^2(\mu)$ that is $\mathcal{F}_T^B$ -measurable there is a unique $\varphi \in H_T$ with*

$$F = \mathbb{E}[F] + \mathcal{I}_T(\varphi) \quad \mu\text{-a.s.}$$

Equivalently, the range of $\mathcal{I}_T$ together with the constants is the whole of $L^2(\mathcal{F}_T^B)$ (`itoIntegralCLM_T_surjective_onto_centered`), and $\mathcal{I}_T$ is an isometric isomorphism of H_T onto the centred part of $L^2(\mathcal{F}_T^B)$ (`itoIsometryEquiv`).

Listing 8. Martingale representation, terminal form (`Foundations/MartingaleRepresentation`).

```
theorem exists_itoIntegral_representation (hBmeas : ∀ t, Measurable (B t))
  (hBcont : ∀ ω, Continuous fun s : ℝ≥0 ↦ B s ω) (T : ℝ≥0)
  (F : Lp ℝ 2 μ)
  (hFmeas : AESTronglyMeasurable[ItoIntegralL2.natFiltration hBmeas T] (↑F) μ) :
  ∃! φ : Lp ℝ 2 (trimMeasure_T (μ := μ) T hBmeas),
    (↑F) ==m[μ] fun ω ↦ (∫ x, F x ∂μ) + itoIntegralCLM_T hB T hBmeas φ ω
```

The process form follows from [Theorem 6.1](#): every square-integrable martingale for the natural filtration is $M_t = M_0 + (\varphi \bullet B)_t$ for $t \leq T$ (`martingale_representation`), because $(\varphi \bullet B)_t$ is the $\mathcal{F}_t$ -conditional expectation of $\mathcal{I}_T(\varphi)$.

The proof uses no Malliavin calculus, no Clark–Ocone formula and no Itô formula for an adapted integrand. The range of $\mathcal{I}_T$ is closed, being the image of an isometry, so it suffices to show that a centred $\xi \in L^2(\mathcal{F}_T^B)$ orthogonal to the range is zero. Two facts do this. First, for every step function h the Doléans exponential $\mathcal{E}(h) = \exp(\int h dB - \frac{1}{2} \int h^2 ds)$ satisfies $\mathcal{E}(h) - 1 \in \text{range } \mathcal{I}_T$ (`stepDoleans_sub_one_mem_range`), starting from the constant step, which is the discounted geometric Brownian motion of [Section 7.7](#), and adding one step at a time with the locality of the integral. Second, an element orthogonal to every $\mathcal{E}(h)$ vanishes (`eq_zero_of_orthogonal_stepDoleans`). Abel summation turns the step exponentials into the exponentials $\exp(\sum_i \lambda_i B_{t_i})$ of finitely many coordinates; on each dyadic cylinder σ -algebra the conditional expectation of an orthogonal element then has a vanishing Laplace transform, which forces it to zero by analytic continuation of the complex moment-generating function and uniqueness of characteristic functions; and by path continuity the dyadic cylinder σ -algebras increase to $\mathcal{F}_T^B$ (`iSup_cylinderFiltration_`

eq_natFiltration), so Lévy's upward theorem sends the conditional expectations to ξ itself. Centring, $\mathbb{E}[\mathcal{I}_T(\varphi)] = 0$, is proved a floor down (integral_itoIntegralCLM_T).

In the parent library this theorem is market completeness: every square-integrable $\mathcal{F}_T^B$ -measurable claim is attained from initial wealth $\mathbb{E}[H]$ by a unique strategy (exists_replicating_strategy). What is not proved is the explicit form of φ , which is the Clark–Ocone formula and needs the Malliavin derivative.

11.2 INTEGRATION AGAINST AN ITÔ INTEGRAL

For $\varphi \in H_T$ let $M = \varphi \bullet B$. The bracket of M is the measure $\langle M \rangle = \varphi^2 \cdot (\lambda_T \otimes \mu)$ on the predictable σ -algebra (bracketMeasure), and the integral against M is defined on $L^2(\langle M \rangle)$ by composing $\mathcal{I}_T$ with the multiplication isometry $\psi \mapsto \psi\varphi$ from $L^2(\langle M \rangle)$ to H_T :

$$\int \psi \, dM = \mathcal{I}_T(\psi\varphi), \quad \left\| \int \psi \, dM \right\|_{L^2(\mu)} = \|\psi\|_{L^2(\langle M \rangle)}$$

(itoIntegralAgainstCLM, norm_itoIntegralAgainstCLM). The chain rule is thus how the object is built; Mathlib has the multiplication isometry only for $p = 1$, so the $p = 2$ case is proved here (LpMulIsometry). What identifies the result as *the* integral against M is the Riemann–Stieltjes value on elementary integrands, $\int Z \mathbf{1}_{(a,b]} \, dM = Z(M_b - M_a)$ for bounded $\mathcal{F}_a$ -measurable Z (itoIntegralAgainst_elementary), together with uniqueness among continuous linear maps with those values (itoIntegralAgainst_unique).

The bracket earns its name. The second moment of an increment is its mass, $\mathbb{E}[(M_b - M_a)^2] = \langle M \rangle((a, b] \times \Omega)$ (norm_sq_increment_eq_bracket); conditionally, $\mathbb{E}[(M_b - M_a)^2 \mid \mathcal{F}_a] = \mathbb{E}[\int_a^b \varphi_u^2 \, du \mid \mathcal{F}_a]$ against the pathwise increment of the bracket (condExp_band_second_moment); and that increment is adapted (bracketProcess_adapted), so $\mathbb{E}[M_b^2 - \langle M \rangle_b \mid \mathcal{F}_a] = M_a^2 - \langle M \rangle_a$ (condExp_sq_sub_bracket): the bracket is the compensator of M^2 . Not formalized are any pathwise quadratic variation of M , a bundled martingale structure for $M^2 - \langle M \rangle$, and the Doob–Meyer decomposition of a general submartingale.

11.3 AGREEMENT WITH THE UPSTREAM CHARACTERISATION

BrownianMotion characterises a stochastic integral axiomatically [2]. A map I from processes to random variables, defined on a domain S , is a *Riemann–Stieltjes extension* if it returns $X(Y_j - Y_i)$ on every elementary process $\mathbf{1}_{(i,j]} X$ with simple $\mathcal{F}_i$ -measurable X , is linear, respects indistinguishability, and passes to the limit under dominated convergence; it is a *stochastic integral* if, in addition, it agrees almost surely with every Riemann–Stieltjes extension on the intersection of their domains.

THEOREM 11.2 (isStochasticIntegral). *For every $\varphi \in H_T$, the integral against $M = \varphi \bullet B$, with integrator M stopped at T , the natural filtration, and as domain the processes indistinguishable from a predictable process square-integrable against $\langle M \rangle$, is a stochastic integral in the sense of BrownianMotion.*

Listing 9. The characterisation (Foundations/StochasticIntegralCharacterisation).

```
theorem isStochasticIntegral (hB : IsPreBrownianReal B μ) :
  IsStochasticIntegral μ (integrator T hBmeas φ hB) (natFiltration hBmeas)
  (sIntegral T hBmeas φ hB) (domain T hBmeas φ)
```

Taking $\varphi = 1$ makes M the Brownian motion stopped at T , so the theorem says in particular that $\mathcal{I}_T$ is a stochastic integral against B in the upstream sense. Two features of the proof are worth recording. The domain is forced rather than chosen: the predicate asks for closure under indistinguishability, and a process indistinguishable from a predictable

one need not be predictable, or even jointly measurable, so the domain is the processes with a predictable square-integrable version, and the integral reads the class of any version. And the uniqueness clause is the substantive part. An extension in the upstream sense is not assumed to be L^2 -continuous, so uniqueness among continuous linear maps does not reach it; the proof is a monotone-class argument from dominated convergence alone, through a Dynkin argument over the predictable rectangles, then simple functions, bounded processes, and finally the whole common domain by truncation. Dominated convergence in L^p , which Mathlib at the pin has only for $p = 1$, is proved on the way for every $p < \infty$ and every measure (`tendsto_eLpNorm_sub_of_dominated_convergence`).

12 WHAT THE CALCULUS CARRIES

The integral and the formula are consumed downstream. Two consumers belong to stochastic analysis proper.

Girsanov's theorem. For a bounded predictable drift θ , $|\theta| \leq C$, let $\frac{dQ}{d\mu} = \exp(-\mathcal{I}_T(\theta) - \frac{1}{2} \int_0^T \theta_s^2 ds)$. Under Q the process $B_t^\theta = B_t + \int_0^t \theta_s ds$ starts at 0, has increments $B_t^\theta - B_s^\theta \sim N(0, t - s)$ for $s \leq t \leq T$, and has jointly independent increments on $[0, T]$ (`Btheta_isQBrownianMotion_predictable_of_bdd`); these fix every finite-dimensional law of B^θ on $[0, T]$ to that of a Brownian motion. Path continuity of B^θ and independence of its increments from the past are not stated. The theorem is the last rung of a ladder (constant, simple adapted, continuous adapted, bounded predictable θ) whose rungs share one lemma: any process supplying its own exponential martingale under Q has the Brownian finite-dimensional laws (`isQBrownianMotion_of_expMartingale`), by characteristic functions, with no adapted-integrand Itô formula. The bounded-predictable statement is made for an adapted modification of B^θ , which has the same finite-dimensional laws. The general case under Novikov's condition is not formalized.

Stochastic differential equations. For Lipschitz b and σ , with constants L_b and L_σ , the Picard map $X \mapsto \eta + \int_0^\cdot b(X) ds + \int_0^\cdot \sigma(X) dB$ is a contraction of H_T with constant $TL_b + \sqrt{T}L_\sigma$, so when that constant is below 1 it has a unique fixed point (`picardMap_exists_unique_fixedPoint`). The diffusion term is the Itô integral of this paper; the fixed point is realized as a sample-path process, $X_t = \eta + \int_0^t b(X_s) ds + (\sigma(X) \bullet B)_t$ with the drift identified as a pathwise Lebesgue integral (`sde_pathwise_decomposition`, `driftContinuousMod_eq_setIntegral`). Pathwise uniqueness of L^2 strong solutions follows from an energy estimate and Grönwall's inequality (`sde_pathwise_uniqueness`); in that statement the diffusion term is an abstract operator whose only assumed property is the Itô isometry bound. Existence on arbitrary horizons, which needs a weighted norm, is not formalized.

13 SCOPE: WHAT IS, AND IS NOT, AN ITÔ CALCULUS HERE

The title claims “a” calculus, and the indefinite article is doing deliberate work. What earns the noun is that the canonical objects and theorems are present and *compose*: the stochastic term of Itô's formula is the isometric integral of [Section 5](#), whose energy law is the isometry of [Section 6](#), whose conditional projection is the martingale of [Theorem 6.1](#), whose continuous representative is the local martingale of [Section 8](#), whose range is characterised by [Theorem 11.1](#), and which is the upstream package's stochastic integral by [Theorem 11.2](#). One object, a single chain of theorems, no re-derivation gluing them. What the indefinite article concedes is equally concrete.

The integrator. The integrator is Brownian motion, or an Itô integral against it. Integration against a general continuous local martingale or semimartingale is not formalized; [Theorem 11.2](#) places the present integral inside the upstream package’s axiomatic characterisation of a stochastic integral.

Square-integrable integrands. Integrands live in L^2 : in H_T on $[0, T]$, or in its unbounded-horizon analogue. The localized class $\{\varphi : \int_0^T \varphi_s^2 ds < \infty \text{ a.s.}\}$ is not covered, which is why the local martingale of [Theorem 9.1](#) is not identified as a stochastic integral.

The scope of each Itô formula. In L^2 , the formula for $f(t, B_t)$ needs the six partial derivatives bounded or of exponential growth in x . In local-martingale form it needs them continuous and nothing else. For Itô processes with adapted coefficients it is proved at a fixed time T , for bounded coefficients with every path continuous, and for f with bounded f' and f'' ; so it does not yet cover e^x or x^2 of such a process, time-dependent f , a random initial value, or coefficients of a stochastic differential equation. The multi-dimensional formula is recorded in the library only as a reduced core, whose hypotheses include its conclusion.

Completeness, not the usual conditions. The pathwise local martingales live on the null-augmented filtration: completeness, which the continuous modification needs, is added, but right-continuity, the other half of the usual conditions, is neither assumed nor used.

Quadratic variation in mean, not pathwise. Quadratic variation is established in L^1 and L^2 along uniform or refining partitions, for Brownian motion and for Itô processes, and the bracket of $\varphi \bullet B$ is identified in conditional mean. No pathwise (almost sure) quadratic variation is formalized, and Lévy’s characterisation of Brownian motion is recorded only as a reduced core.

None of these boundaries qualifies a statement made in [Sections 5 to 12](#); they bound the *reach* of the development, not the *validity* of any theorem in it.

14 RELATED WORK

Keskin’s Isabelle/HOL martingale library [\[10\]](#) is the closest prior art in the probabilistic direction; it develops sub- and supermartingales in discrete and continuous time and stops below stochastic integration. Degenne, Ledvinka, Marion and Pfaffelhuber [\[5, 2\]](#) construct Brownian motion in Lean, through the Kolmogorov extension theorem, Gaussian measures and a Kolmogorov–Chentsov theorem; their package is the foundation of this work, and its stochastic-integration programme (an axiomatic characterisation of the integral, local martingales, quadratic variation of locally square-integrable martingales, and the Doob–Meyer decomposition) is complementary to it: a general theory under construction, against which [Theorem 11.2](#) certifies the concrete Brownian integral built here. Nagy’s Lean derivation from Itô’s lemma to Black–Scholes [\[12\]](#) formalizes the structural drift computation of Itô’s lemma and, on its own account, leaves the continuous Itô integral structurally verified rather than constructed. Avigad, Hölzl and Serafin’s central limit theorem [\[1\]](#) remains the historical anchor for deep measure-theoretic probability in proof assistants, and Echenim, Guiol and Peltier formalized discrete-time replication pricing in Isabelle/HOL [\[7\]](#). StochBench [\[13\]](#) is a recent Lean benchmark of graduate stochastic-process problems which, lacking infrastructure such as the Itô integral, states many of its targets with the required properties as hypotheses. A search in October 2026 of the Isabelle Archive of Formal Proofs and the Lean and Coq/Rocq ecosystems found no other

completed machine-checked construction of the Itô integral, proof of Itô's formula, or martingale representation theorem.

15 CONCLUSION

The development formalizes the Itô calculus of Brownian motion: on $[0, T]$ the integral as an isometry, the integral as a continuous L^2 martingale with its full second-moment structure, its range (martingale representation) and its uniqueness (agreement with the upstream axiomatic integral); Itô's formula in L^2 under bounded or exponentially growing derivatives, as a process, and for Itô processes with adapted coefficients; and the pathwise object, a continuous local martingale on the half-line, to which Itô's formula extends without any growth condition. To our knowledge it is the first machine-checked Itô calculus in any proof assistant. It is also the layer the parent library [4] stands on, where the integral and the formula are consumed to construct, rather than assume, objects of continuous-time pricing: the dynamic change of measure, the replicating hedge, and the discounted price as an Itô integral. The boundaries of [Section 13](#) (general semimartingale integrators, the localized integrand class, pathwise quadratic variation, and the multi-dimensional formula) mark where the formalized theory currently ends.

Provenance. The Lean development was written by the author with an AI coding assistant (Claude Code), as recorded in the repository's provenance file; every proof is checked by the Lean kernel and passes the gates of [Section 2](#) unchanged. This manuscript was revised with the same assistant, and its statements were checked against the Lean sources of release v1.5.0.

A DISPLAYED RESULTS AND THEIR LEAN NAMES

Result	Lean name	Module / benchmark entry
Rectangle π -system	isPiSystem_predictableRect	ItoIntegralCLM
Generation	generateFrom_predictableRect	ItoIntegralCLM
Density	simpleAssembly_T_denseRange	ItoIntegralCLM
The Itô integral, isometry	itoIntegralCLM_T, itoIntegralCLM_T_norm	ItoIntegralCLM
Unbounded horizon	itoIntegralL2_norm	ItoIntegralL2Dense
Keystone $\mathcal{I}_T(g_B)$	itoIntegralCLM_T_brownian	ItoIntegralBrownian
Conditional increments	condExp_adapted_mul_increment	ItoIntegralProcessMartingale
Key identity (Theorem 6.1)	itoProcessCLM_eq_condExpL2	ItoIntegralProcessGeneral
Martingale property	itoIntegralProcessGen_isMartingale	ItoIntegralProcessGeneral
Time-indexed isometry	itoProcessCLM_norm_sq	ItoIntegralProcessIsometry / sc-ito-general-time-isometry
QV in L^2 , in L^1	tendsto_qv, qv_equals_t	QuadraticVariationL2, BrownianQuadraticVariation / sc-thm-6.1.1
Weighted QV	tendsto_weighted_qv_process	WeightedQuadraticVariation
Itô, bounded (Theorem 7.1)	ito_formula_L2_bddDeriv	ItoFormulaCLM / sc-thm-7.1.1
Itô, time-dependent (Theorem 7.2)	ito_formula_td_L2_bddDeriv	ItoFormulaTD / sc-thm-7.1.2
Itô as a process	ito_formula_td_process	ItoFormulaProcess / sc-ito-formula-td-process
Itô, exponential growth (Theorem 7.3)	ito_formula_td_localized	ItoFormulaLocalized / sc-ito-formula-localized
GBM; discounted GBM	ito_formula_gbm, discountedGBM_eq_itoIntegral	ItoFormulaGBM / sc-ito-formula-gbm, sc-discounted-gbm-ito
Expectation Itô	expectation_ito	FeynmanKacHeatEquation
Continuous modification	exists_continuous_modification_ itoProcess	ItoIntegralProcessContinuousModification
Local martingale on $[0, T]$	exists_continuous_localMartingale_ modification	ItoIntegralProcessLocalMartingaleGeneral
Local martingale on $\mathbb{R}_{\geq 0}$	exists_continuous_localMartingale_ modification_infinite	ItoIntegralProcessLocalMartingaleInfinite / sc-ito-infinite-local-martingale
Itô, no growth (Theorem 9.1)	ito_formula_unrestricted	ItoFormulaUnrestrictedLocMart / sc-ito- formula-unrestricted-islocalmartingale
Exit times	exitTime	ExitTime
Itô, constant coefficients	ito_formula_itoProcess	ItoFormulaItoProcess / sc-ito-formula-ito-process
Itô, adapted coefficients (Theorem 10.1)	ito_formula_adapted	ItoFormulaAdapted / sc-ito-formula-adapted
Representation (Theorem 11.1)	exists_itoIntegral_representation, itoIntegralCLM_T_surjective_onto_ centered, itoIsometryEquiv	MartingaleRepresentation
Representation, process form	martingale_representation	MartingaleRepresentation / gir-thm-9.3.4
Integral against $\varphi \bullet B$	itoIntegralAgainstCLM, itoIntegralAgainst_elementary	ItoIntegralAgainstMartingale / sc-ito-integral-band
Bracket, conditional Characterisation (Theorem 11.2)	condExp_band_second_moment isStochasticIntegral	PointwiseBracket / sc-bracket-conditional StochasticIntegralCharacterisation / sc-ito-is-stochastic-integral
Girsanov, bounded predictable	Btheta_isQBrownianMotion_ predictable_of_bdd	GirsanovPredictableTheta / gir-thm-9.1.8-predictable
SDE existence and uniqueness	picardMap_exists_unique_fixedPoint	SDEExistence / sde-picard-existence-uniqueness

All modules are under MathFin/Foundations/. The library contains no sorry. Every theorem in the table is pinned directly by one of the two axiom audits to the three standard axioms, except the supporting lemma isPiSystem_predictableRect, which is used in the proofs of pinned theorems; definitions are not pinned separately.

REFERENCES

- [1] J. Avigad, J. Hölzl, L. Serafin. A formally verified proof of the central limit theorem. *Journal of Automated Reasoning*, 59(4):389–423, 2017.
- [2] R. Degenne et al. *brownian-motion*: a Lean 4 development of Brownian motion and stochastic integrals. <https://github.com/RemyDegenne/brownian-motion>, 2024–2026. Consumed at commit 0d5b6eb.
- [3] R. Coelho. The fundamental theorem of asset pricing, formalized in Lean 4. arXiv:2606.28990, 2026.
- [4] R. Coelho. A formally verified library of mathematical finance in Lean 4. arXiv:2606.01356, 2026.
- [5] R. Degenne, D. Ledvinka, E. Marion, P. Pfaffelhuber. Formalization of Brownian motion in Lean. arXiv:2511.20118, 2025.
- [6] L. de Moura, S. Ullrich. The Lean 4 theorem prover and programming language. In *CADE 28*, LNCS 12699, pp. 625–635, 2021.
- [7] M. Echenim, H. Guiol, N. Peltier. Formalizing the Cox–Ross–Rubinstein pricing of European derivatives in Isabelle/HOL. *Journal of Automated Reasoning*, 64:737–765, 2020.
- [8] K. Itô. Stochastic integral. *Proceedings of the Imperial Academy, Tokyo*, 20(8):519–524, 1944.
- [9] I. Karatzas, S. E. Shreve. *Brownian Motion and Stochastic Calculus*, 2nd ed. Springer, 1991.
- [10] A. Keskin. A formalization of martingales in Isabelle/HOL. arXiv:2311.06188, 2023.
- [11] The mathlib Community. The Lean mathematical library. In *CPP 2020*, pp. 367–381, 2020.
- [12] T. Nagy. From Itô to Black–Scholes: a machine-verified derivation in Lean 4. SSRN Working Paper 6336503, 2026.
- [13] I. Davidovich, D. Ganguly, V. Singh, V. Chaudhary. StochBench: a domain-specific benchmark for stochastic processes in Lean. arXiv:2609.09264, 2026.